Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/26/2005
PI Name:	Lee, David M. Ph.D.		
Project Title:	Studies of Atomic Free Radicals in a Cryogenic Environment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dml20@cornell.edu	**Fax:**	FY 607-255-6428
PI Organization Type:	UNIVERSITY	**Phone:**	607-255-5286
Organization Name:	Cornell		
PI Address 1:	Physics Department, Laboratory of Atomic and Solid State		
PI Address 2:	Physics, Cornell University		
PI Web Page:	http://www.lassp.cornell.edu/		
City:	Ithaca	**State:**	NY
Zip Code:	14853	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	04/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Lee, David M.		
COI Name (Institution):			
Grant/Contract No.:	NAG 3-2871		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity		

A wide range of techniques have been applied to the study of Impurity helium Solids, which consist of impurity atoms or molecules or clusters of these impurities surrounded by thin layers of solid helium. These solids are porous and the pores can be filled with superfluid 4He. Ultrasound, X ray diffraction, nuclear magnetic resonance and electron spin resonance have all been applied to determine the nature of these substances. Studies of low temperature chemical reactions, studies of potential quantum fluid behavior and studies to maximize energy storage are all under way. Impurity-Helium (Im-He) solids are truly fascinating substances. An important goal of this research program is to investigate these substance by a variety of experimental techniques in order to advance our knowledge of quantum solids, low temperature chemical reactions and complex materials consisting of nanoclusters of impurities embedded in a mixture of liquid and solid helium. The criterion for quantum liquid and solid behavior is that the thermal de Broglie

Task Description:

wave length is comparable to or greater than the mean interparticle spacing. Of course the liquid helium contained in the pores of these substances is a true quantum fluid. Clusters of H2 or D2 molecules embedded in Im-He solids are quantum solids. On the other hand, the heavier impurities behave classically. Atomic hydrogen is found to be quite mobile in impurity-helium solids. If low enough temperatures and high enough densities are achieved, it might be possible to observe Bos-Einstein condensation which would be a truly exciting discovery! The assembly of hydrogen atoms would then constitute a separate new quantum fluid.
Chemical reactions such as the recombination of H atoms into H2 molecules are spontaneous and do not require thermal activation. Chemical reactions involving encounters between H atoms and H2 molecules, for example, might be expected to be thermally activated. Nevertheless, even at temperatures in near 1.3 K chemical reactions can proceed. The reaction mechanism involves quantum mechanical tunneling and the reactions are called exchange tunneling reactions. Studies of the kinetics of the reaction involving D atoms and H2 molecules yielding HD molecules and H atoms have been performed. This reaction is followed by a slower reaction involving D atoms and HD molecules, yielding H atoms and D2 molecules. Very high concentrations of H atoms can be achieved via exchange tunneling reactions. Attempts are still underway to maximize the yield of H atoms to provide high enough concentration to get into quantum fluid territory. Electron spin resonance experiments have been employed and will continue to be employed in this work to determine H atom and D atom concentrations.

It is important to obtain further details regarding the structure of impurity-helium solids. To this end, x ray scattering and electron spin resonance methods are being employed. Analysis of the x ray diffraction patterns indicate that the impurity atoms tend to form small (nanoscale) clusters. (Each cluster is thought to be surrounded by a very thin layer of solid helium.) As the sample is warmed above the superfluid 4He transition, the clusters grow. New x ray experiments are planned to investigate this clustering phenomenon in more detail and to search for the possible presence of isolated impurity atoms or molecules.

The environment surrounding impurity atoms in the clusters may be obtained by observing the detailed structure of the electron spin resonance signals. Satellite lines on the signals indicate that the microwave transitions tend to induce spin flips of hydrogen nuclei on neighboring H2 molecules. Higher magnetic fields will allow satellite lines corresponding to deuteron spin flips to be observed. Such experiments are currently being planned. So far it has not been possible to observe isolated impurity atoms with no nearest neighbor impurities in our samples. Sensitive electron spin resonance experiments will be conducted to search for these isolated atoms which are expected to have very long relaxation times.

Experiments are being continued to maximize the concentration of nitrogen atoms in impurity helium solids. The nitrogen atom recombination reaction is one of the most energetic chemical reactions, so an impurity helium solid containing high concentrations of nitrogen atoms would provide a very effective energy storage medium. One interesting phenomenon at very high atom concentrations is exchange narrowing. As the concentration increases on might expect that the increasing dipole-dipole interactions would lead to increasing line widths. If the interatomic spacing becomes small enough, however, a drastic line narrowing associated with exchange effects would set in with further concentration increases.

Finally, for the case of impurity-helium solids with free radicals, the presence of unpaired spin leads to the possibility that magnetic forces can suspend a sample, thereby simulating microgravity conditions. This phenomenon has been observed in the course or our research program. Magnetic fields can also be used to transfer impurity helium solids containing paramagnetic atoms between different locations. Impurity-helium solids provide a unique opportunity to study a wide variety of fundamental phenomena as well as to investigate possible practical applications. Impurities may be varied from the heaviest atoms or molecules which behave classically to the lightest atoms or molecules which can give rise to quantum phenomena. The concentration of impurities may also be varied at will. The Im-He solids are porous solids which can be studies with the pores filled with liquid helium or as free standing solids with empty pores. Studies of impurity-helium solids can give significant insight into the nature of porous media in general. The structures of these unique substances bear close resemblance to the light aerogels.

For the case of Im-He solids containing the lightest impurities, it might be possible for the ensemble of impurities to act as a quantum fluid or solid. This situation will arise only if the thermal de Broglie wave length is comparable or greater than the mean interparticle spacing between the impurity atoms. Interesting phase transitions might be observed such as Bose-Einstein condensation of hydrogen atoms. Exchange tunneling reactions have also been studied. These reactions can take place in the low temperature regime where thermally activated chemical reactions can no longer occur. They have been used to achieve very high concentrations of atomic hydrogen which ultimately allow entry into the quantum regime. These tunneling reactions may also provide a unique transport mechanism for hydrogen atoms in solid H2. In the next round of experiments, such transport processes will be investigated.

Matrix isolated active free radicals have been considered for possible energy storage application for many years. The impurity-helium solids provide a medium in which very large concentrations of reactive atoms can be achieved and stored. The chemical reactions corresponding to the recombination of these atoms into molecules are highly exothermic. Future applications to special explosives and rocket fuels may ultimately be possible.

Finally, tunneling reactions involving isolated reactive atoms may also occur on interstellar dust particles.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress for the Current Year:
The results of research performed in the previous year are listed in the bibliography. The first item in the bibliography is an archival paper entitled “Deuterium Atoms and Molecules in Nanoclusters of Molecular Deuterium” which appeared in the Physical Review B in March 2004. In this work, x-ray diffraction and CW electron spin resonance (ESR) techniques were employed to investigate the structure and behavior of impurity-helium solids created by introducing deuterium atoms and molecules into superfluid 4He. Mixed D2-Ne-He samples were also investigated in this work for the purpose of comparing the sizes and the behavior of clusters created when neon and deuterium were simultaneously present in the make-up gas mixture. X-ray diffraction experiments revealed that the D2-He and D2-Ne-He samples have characteristic impurity densities of order E 21 per cubic centimeter. The impurities were found to be mainly present in small clusters each surrounded by thin layers of solid helium 4. The diameters of the impurity clusters were typically in the range 50-90 angstroms. Samples containing both atomic and molecular deuterium were prepared by passing the

Task Progress:	make-up gas consisting of a mixtures of helium 4 and molecular deuterium through a high power radiofrequency discharge to achieve dissociation of the molecular deuterium. The atomic deuterium in the samples was directly investigated via ESR, which allowed the ground state spectroscopic parameters, line shapes, relaxation times and saturation behavior to be studied. The results of the x-ray and ESR measurements were complementary, and provided the means to characterize the properties of impurity-helium solids created with light impurities. By means of ESR, the average concentration of deuterium atoms in samples containing atomic and molecular deuterium impurities was determined to be 1.5 E 18 per cubic centimeter. The local concentration of deuterium atoms within a given cluster was estimated to of order 2 E 19 per cubic centimeter for typical clusters with diameters in the range 50-90 angstroms (as obtained from the x-ray diffraction studies). This density of deuterium atoms in the clusters is sufficient to satisfy the quantum fluid criterion for temperature less than about 175 mK for a gas of free deuterium atoms. Unfortunately the deuterium atoms in the clusters are not free but appear to be trapped at lattice sites as manifested by the very slow recombination rate of deuterium atoms into deuterium molecules. Two potential applications of impurity-helium solids containing deuterium are as follows: 1. Very cold (1 mK) deuterium-helium solids should be able to act as extremely efficient moderators, making it possible to make one hundred fold improvements in the available flux of cold neutrons. 2. Deuterium-helium solids cooled to 3 mK in a very high magnetic field ~ 10T will have very high nuclear polarizations. Spin polarized deuterium nuclei can be expected to enhance the efficiency of nuclear fusion reactions involving deuterons. The second item in the bibliography is the abstract for a paper entitled “Long term stability of H atoms in HD-D2-helium solids”. This paper was presented at the Quantum Fluids and Solids Conference held in Trento, Italy in the summer of 2004 and will be published in the Journal of Low Temperature Physics in early 2005. In this work, a mixed sample containing atomic hydrogen, atomic deuterium, molecular hydrogen and molecular deuterium was prepared. Exchange tunneling reactions then took place which led to a decrease in the atomic deuterium concentration in the sample. For a sample formed from a gaseous mixture containing hydrogen molecules, deuterium molecules and helium atoms in the ratio 1:4:100, the hydrogen atom concentration was essentially constant. For this particular gas mixture, the rate of evolution of hydrogen atoms produced in the exchange tunneling reaction between deuterium atoms and molecular hydrogen or HD almost exactly balanced the loss of hydrogen atoms via molecular recombination. The experiment lasted over 40 hours and during this time the hydrogen atom concentration was nearly unchanged. The local concentrations of hydrogen atoms in typical clusters were estimated to be ~ 1.5 E 20 per cubic centimeter. Satellite ESR lines associated with simultaneous spin flips of hydrogen atoms and protons on neighboring hydrogen or HD molecules were also studied in this work. The relative intensities of the main line and the satellite lines were used to estimate the separation between the atomic free radical hydrogen or deuterium atoms and the protons forming neighboring hydrogen molecules by means of the Trammel formula. This separation we estimated to be about 2.1 angstroms. The main virtue of this work is that it demonstrates the possibility of maintaining a large, relatively stable population of hydrogen atoms for long time periods. Once the supply of deuterium atoms is exhausted by the exchange tunneling reactions, hydrogen atoms will no longer be produced so one would expect that the hydrogen atom concentration would begin to decrease due to molecular recombination. The time required for this trend to be observed is well over 40 hours. Finally, for the past year, a major project has been underway to design and construct a pulsed ESR spectrometer. The centerpiece of this apparatus is a traveling wave tube amplifier which is needed to supply intense short (~ 40 nanosecond) microwave pulses to provide large tipping angles required for spin-echo experiments. It was necessary to reduce the Q of the TE011 resonant cavity to ~ 700 in order to shorten the ring down time, thereby minimizing interference between the cavity ring down and the free induction decay signal. The construction is now complete and experiments are presently underway with this spectrometer. The first experiments were performed on an impurity-helium solid containing large concentrations of N atoms and molecules. The high density of atomic nitrogen in these samples led to very short free induction decay times, corresponding to large ESR line widths associated with dipolar broadening. Because of these large line widths (~ 50 gauss), only a small portion of the nitrogen atoms could be excited since the minimum pulse length for a 90 degree pulse is only 40 nanoseconds as determined by the maximum power provided by the traveling wave tube amplifier. A shorter higher power pulse would allow a larger portion of the ESR line to be encompassed. In spite of these difficulties, spin echo signals were observed in impurity-helium solids containing large concentrations of atomic nitrogen. The spin echoes were small as would be expected because only a small portion of the nitrogen atoms were excited by the pulses. For 90 – t – 180 degree pulse sequences, the echoes rapidly diminished in amplitude at t increased. This effect is probably attributable to spectral diffusion which is a result of strong dipole-dipole interactions between the magnetic moments of the nitrogen atoms. Recently, experiments were performed on impurity-helium solids containing only deuterium atoms and molecules. The smaller concentrations of atomic deuterium led to much smaller dipolar broadening. For this case, the lines were sufficiently narrow to be completely excited by the applied pulses. Very large echoes were observed for this case. Similar results were obtained for hydrogen atom impurities.
Bibliography Type:	Description: (Last Updated: 03/20/2008)
Articles in Peer-reviewed Journals	Bernard, E. P., Boltnev, R. E., Khmelenko, V. V., Kiryukhin, V., Kiselev, S. I., , and Lee, D. M. "Deuterium atoms and molecules in nanoclusters of molecular deuterium" PACS number(s): Phys. Rev. B 67.40. Yv, 67.40. Mj, 62.65. +k , Mar-2004
Papers from Meeting Proceedings	Bernard, E.P.; Boltnev, R.E.; Khmelenko,V.V.; and Lee, D.M. "Long term stability of H atoms in HD-D2-He solids" Quantum Solids and Liquids Conference, Trento, Italy 2004 (to be published in J. Low Temperature Physics) Feb-2005